ELECTROCHEMISTRY

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

Some metals and their salt solutions are given.

Zn, Fe, Ag, Cu, Mg, $ZnSO_4$ solution, $FeSO_4$ solution, $CuSO_4$ solution, $MgSO_4$ solution, $AgNO_3$ solution

[Hint : Reactivity : Mg > Zn > Fe > Cu > Ag]

(a) Which of the given metals are used to make a galvanic cell with highest cell voltage ?

(b) Write the chemical equation of the reaction taking place at the cathode in this cell.

(c) Which of the given metals acts only as the anode when galvanic cells are made ?

2. 2026 **2 MARKS**

Two activities are given.

(i) Ag rod is dipped in $ZnSO_4$ solution.

(ii) Fe rod is dipped in $CuSO_4$ solution.

(a) In which of these solutions does the displacement reaction take place ?

(b) Which metal undergoes oxidation in the above reaction ?

3. 2026 **1 MARK**

Assertion (A) : When electricity is passed through an aqueous solution of sodium chloride, sodium is obtained at the cathode.

Reason (R) : The reduction tendency of Na^+ is higher than that of H_2O.

(a) Both (A) and (R) are correct, and (R) is the correct explanation of (A).

(b) Both (A) and (R) are correct, and (R) is not the correct explanation of (A).

(c) (A) is correct, but (R) is not correct.

(d) Both (A) and (R) are not correct.

4. 2025 **4 MARKS**

Three metals Ag, Mg, Cu and their salt solutions are given.

(a) How many galvanic cells can be constructed using these metals ?

(b) If we construct a galvanic cell using the most reactive metal and the least reactive metal from the given metals, identify the cathode and anode.

(c) Write the chemical equation of the reaction which takes place at anode and cathode in the above cell.

5. 2025 **3 MARKS**

Copper is electroplated on an iron bangle.

(a) Which electrolyte is used here ?

(b) Write any one of the advantages of electroplating.

(c) Does the intensity of colour of electrolyte change during this process ? Why ?

6. 2024 **4 MARKS**

The order of reactivity of some elements are given. Analyse the following questions.

Mg > Zn > Fe > Cu > Ag

(a) Name any one metal that cannot displace hydrogen from dilute hydrochloric acid.

(b) Which are the metals that can displace Fe from $FeSO_4$ solution ?

(c) A galvanic cell is constructed using Zn and Fe as electrodes.

(i) Which is the anode of this cell ?

(ii) Write the equation of the chemical reaction taking place at the cathode.

7. 2024 **3 MARKS**

Silver is coated on copper bangle by electroplating.

(a) Which metal is connected to negative terminal of the battery in this process ?

(b) Which is the electrolyte used here ?

(c) Write the chemical equation of the reaction that takes place at the positive electrode.

8. 2024 **1 MARK**

Which is the product obtained at the cathode when sodium chloride solution is electrolysed ?

(Na, Cl_2, O_2, H_2)

9. 2022 **5 MARKS**

A picture of galvanic cell is given below :

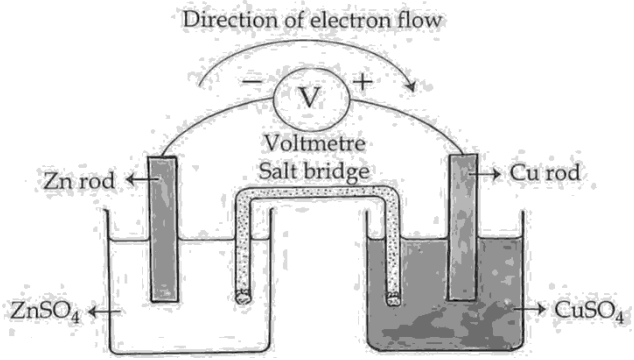
(a) What is the energy change taking place in a galvanic cell ?

(b) Identify the anode in the given cell.

(c) Write the chemical equation of the reaction taking place at anode.

(d) In which electrode does oxidation take place ?

(e) Write the chemical equation of the redox reaction in the cell.



10. 2022 **2 MARKS**

(a) What is electroplating ?

(b) Which is the electrolyte used in electroplating of copper on an iron bangle ?

11. 2022 **1 MARK**

Which metal is deposited at cathode, when molten sodium chloride is electrolysed ?

(Sodium, Hydrogen, Chlorine, Calcium)

12. 2022 **1 MARK**

Which gas is produced when metals react with dilute hydrochloric acid ?

13. 2021 **4 MARKS**

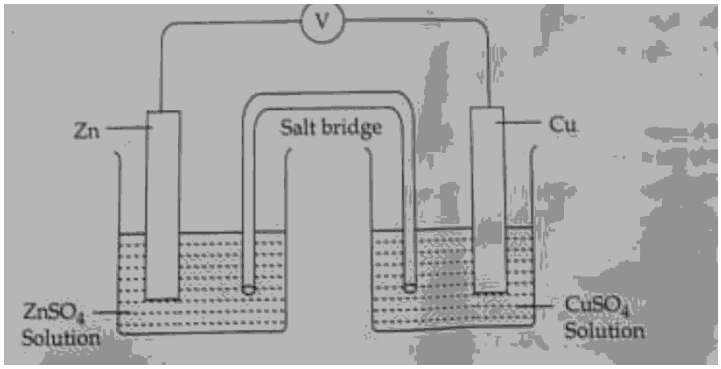
The figure of a galvanic cell is given.

[Hint : Zinc is more reactive than Copper]

(a) Identify anode and cathode.

(b) What will be the direction of flow of electrons ?

(c) Write the chemical equation of the reaction taking place at anode.



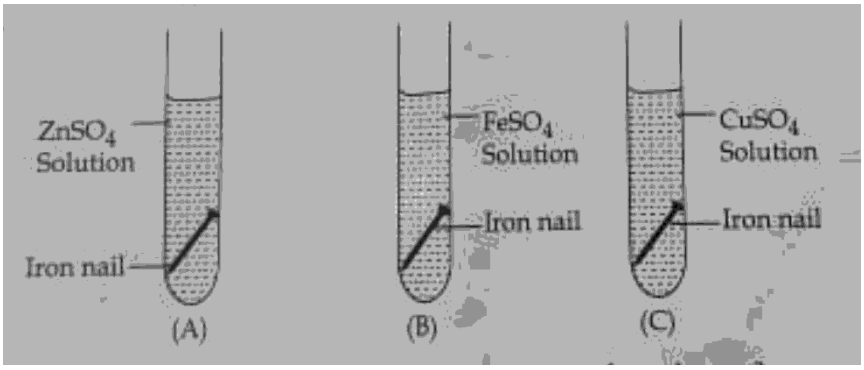
14. 2021 **3 MARKS**

Analyse the figure and answer the following questions.

(a) In which test tube does the iron nail undergo a colour change ?

(b) Write the name of the reaction taking place here.

(c) Write the chemical equation of this reaction.



15. 2021 **2 MARKS**

Some metals are given in the box. Fe Mg Cu Pb Zn

(a) Which metal among these reacts vigorously with dilute HCl ?

(b) Which gas is liberated by this reaction ?

16. 2021 **2 MARKS**

(a) When molten Sodium Chloride is electrolysed, which is the gas liberated at the anode ?

(b) Write the chemical equation of the reaction taking place at the cathode.

17. 2021 **1 MARK**

Which is the electrolyte used to electroplate Silver on an iron bangle ?

18. 2021 **1 MARK**

What is the energy change that takes place in galvanic cell ?